EXPERIMENT E7 · VARIANT C · MEASUREMENT REVIEW → REVISION B

# Will the deformation measurement work?

27 Oct 2026 · read as the person who will run the tests · numbers from the axisymmetric heat model (sim_thaw.py), indicative

**Verdict.** The rig will run, and temperature, TDR and Pt100 will be measured without trouble. Deformation is the delicate part: in the planned unsaturated sand the expected heave is about as large as the thermal drift of the rig itself. Revision B removes most of the rig-side problems. The sample side needs a decision before buying.

## Findings and what revision B changes

| TOPIC | PROBLEM IN REV. A | CHANGE IN REV. B | EFFECT | STATUS |
|---|---|---|---|---|
| **Signal size** | Sr ≈ 0.39: ice adds 1.3 % volume, pore air is 23 % → in a closed sample the expansion fits into the pores | none on the rig; see decision below | expected in-situ heave ≈ 0 (tens of µm at most); upper bound 1.35 mm | ▲ decide |
| **Rig drift** | LVDT on the Al beam with a steel rod; camera referenced to the PETG rim | Invar 36 post from the base plate carries the LVDT and a camera target; Invar guide rod | per 20 K: LVDT −24…+48 → −59…0 µm, left after blank-run correction 12 → ≤ 6 µm; camera 0…100 → −49…0 µm, left 8 → ≤ 3 µm | ✓ fixed |
| **Freezing direction** | cell on a 25 mm Al plate freezes from below and the sides too; last unfrozen spot 27 mm above the bottom | cell printed in one piece ⌀240 × 200: +40 mm low-infill wall and base (λ ≈ 0.06), 3 G10 pads in base pockets; beam gap 200 → 160 keeps the height (598 mm) | freezes from the top: last unfrozen 13 mm above the bottom; 95 % frozen in 13 h (was 9 h) | ✓ fixed |
| **Laser** | 45° laser: collar shadows a 25 mm strip, 41° DOE gives only 7–8 lines | laser at 68° on a bracket under the beam, 124 mm from the centre | no shadow, 11 lines at 9.6 mm; line shift 0.40 Δh | ✓ fixed |
| **Camera** | CSI ribbon cannot pass the lid seal; separate heater near the sample | RPi 5 inside the camera box (its ≈ 5 W keep the window clear); CR1000 C5 pulse triggers the photo; 6 mm lens | Ethernet + 5 V through the seal only; same time stamps without NTP | ✓ fixed |
| **Bearing** | LM12UU ball bushing: grease stiffens at −10 °C, ice at thaw → stick-slip, load not constant | igus drylin RJ4JP-01-12, dry polymer, same size | load change from friction < 0.02 kPa | ✓ fixed |
| **Moisture** | open surface sublimates while frozen; frost and dew at the start of thaw look like heave | weigh the cell before and after each cycle; SHT45 air RH on the RPi; optional window cover over the collar | loss and frost become visible in the data | ✓ added |
| **Freezer air** | on/off control swings ±2–3 K; air sensor only optional | 80 mm fan; Campbell 109 air probe required | boundary condition known; smaller drift noise | ✓ added |

### ▲ Decision needed before buying

What should E7 measure?

- **Frost heave:** raise saturation to Sr ≥ 0.85, or supply water from below (open system). Then heave of 0.1–1 mm is realistic.
- **Thaw settlement under load at Sr 0.3–0.5:** keep the sample; expect tens of µm and rely on the blank-run correction.

### Before the first real cycle

1. Blank run: Al dummy instead of sand, one full cycle; fit drift against cell temperature.
2. Calibrate the camera with shims 0.1 / 0.5 / 1.0 mm under a flat plate.
3. Measure the freezer: rig 598 mm vs ≈ 650 mm inside.
4. Confirm LVDT operation at −10 °C with RDP / Elkome.
5. Program: 24 h at −10 °C, 24 h at +10 °C (thaw takes ≈ 13 h).

## Cost and resolution

**≈ 1,450–1,980 €**
rev. B excl. VAT (rev. A 1,365–1,870 €); frame from laser-cut plates and printed parts, no machining

**≤ 6 µm**
LVDT drift left after the blank-run correction (model), LVDT resolution ≈ 1 µm

**≈ 10 µm**
camera height map per laser line after averaging ≈ 1000 px along the line

Details: drawings E7-V01…V05 rev. B, purchase list, calculations §15 (vypocty.md), 3D model with drift chart: E7 sample thawing.

Model assumptions: θ 0.15, n 0.38, λ sand 1.2 / 1.9 W/mK, h = 7 W/m²K in still air; frame parts as lumped elements. Treat all numbers as estimates to be checked by the blank run.